

Intraday Volume–Price Dynamics in Semiconductor Stocks

AN ANALYSIS OF TIME-OF-DAY VOLATILITY AND PRICE IMPACT

BAIM 635

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Research Question

What is the relationship between trading volume and short-term price movements in semiconductor stocks, and does price impact vary across companies?

Tools Used:

- Postgresql – Gather all Datasets of semiconductors and ETF
- SAP Expert Analytics – Calculated columns
- Excel – Column Charts
- R studio – Column Types, Aggregation, Heatmaps
- Python – Regression and Correlation

Data Summary

- 20 Tickers : 19 Companies, 1 ETF (SOXX)
- Data Extracted from PgAdmin :
polygon_1m_bars
- Data Content : 3 years 2021 - 2023
- Sql Code available in the Appendix Section
- Columns of each Ticker :

Ticker Semiconductor	Company Name
NVDA	NVIDIA Corporation
AMD	Advanced Micro Devices, Inc.
AVGO	Broadcom Inc.
QCOM	Qualcomm Incorporated
TXN	Texas Instruments Incorporated
INTC	Intel Corporation
MCHP	Microchip Technology Incorporated
LRCX	Lam Research Corporation
KLAC	KLA Corporation
MU	Micron Technology, Inc.
SWKS	Skyworks Solutions, Inc.
QRVO	Qorvo, Inc.
MPWR	Monolithic Power Systems, Inc.
NXPI	NXP Semiconductors N.V.
SOXX	
AMAT	Applied Materials, Inc.
TER	Teradyne, Inc.
IPGP	IPG Photonics Corporation
WDC	Western Digital Corporation
STX	Seagate Technology

tickerdate	tickertime	open	high	low	close	vw_price	volume	trades
1/4/2021	9:30:00	524.17	524.2	518.99	519.13	522.6221	208089	2936

INSIGHT-1 Volume–Volatility Relationship

TOP 3 STOCKS (Highest Expected Volume–Volatility Sensitivity)

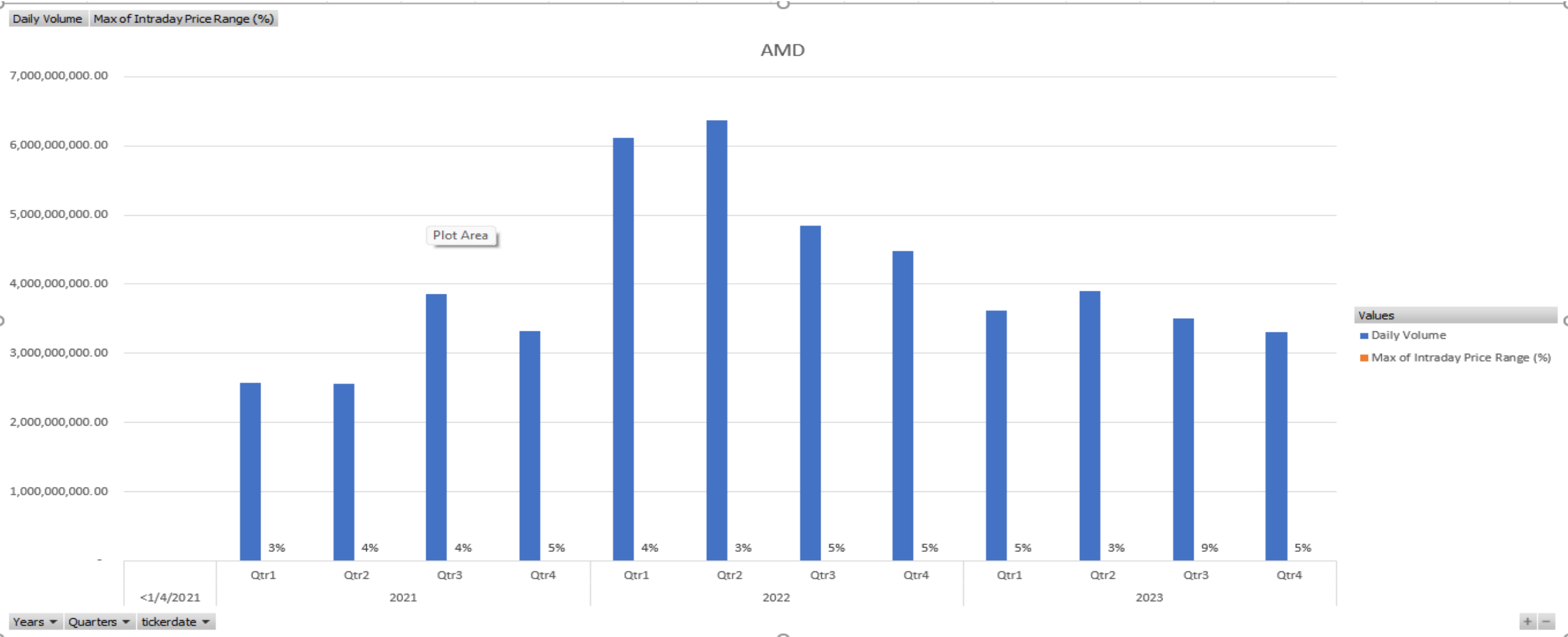
- 1) AMD**
- 2) NVIDIA**
- 3) MU**

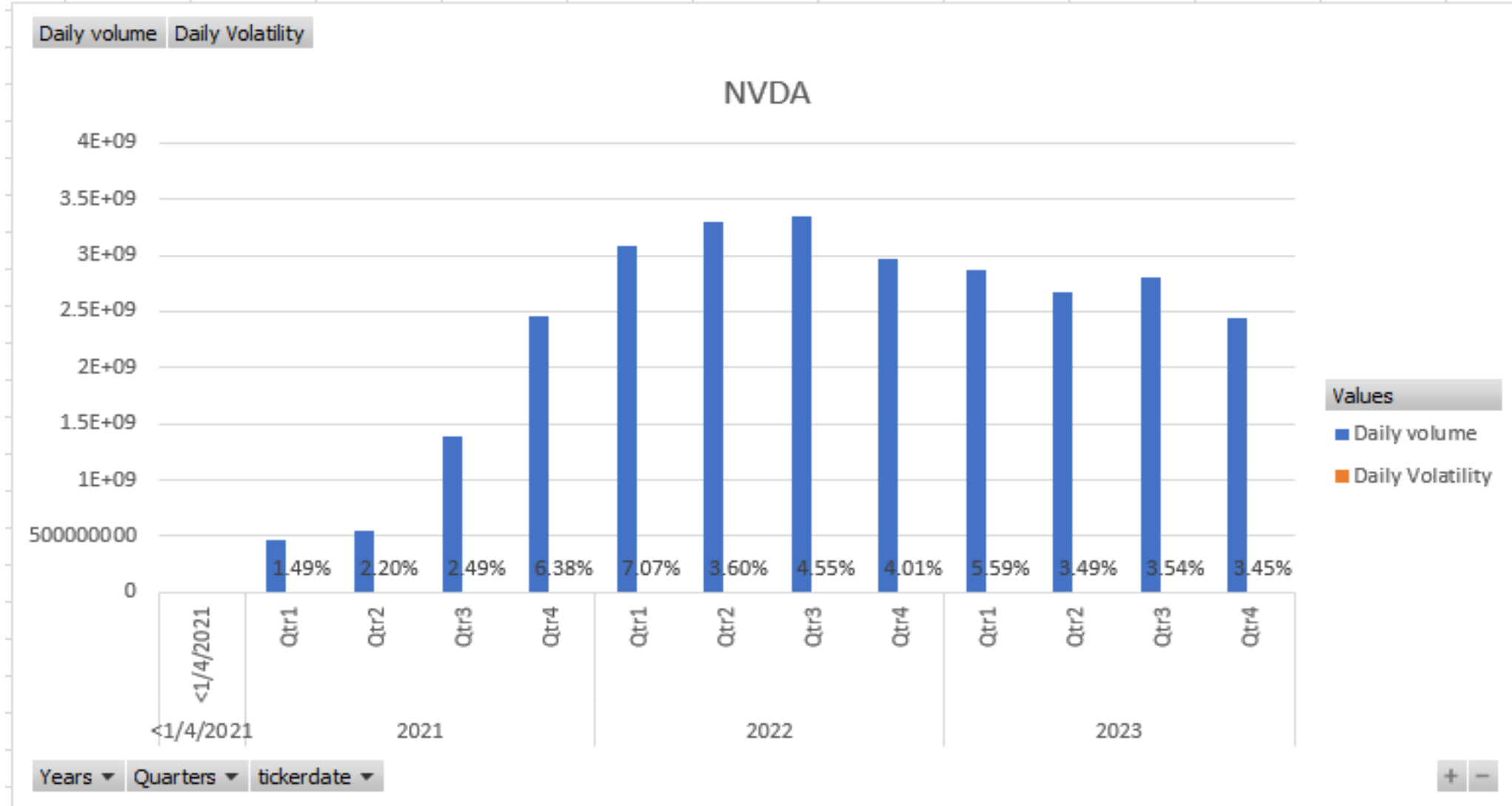
BOTTOM 3 STOCKS (Lowest Expected Volume–Volatility Sensitivity)

- 1) SOXX**
- 2) TXN**
- 3) AVGO**

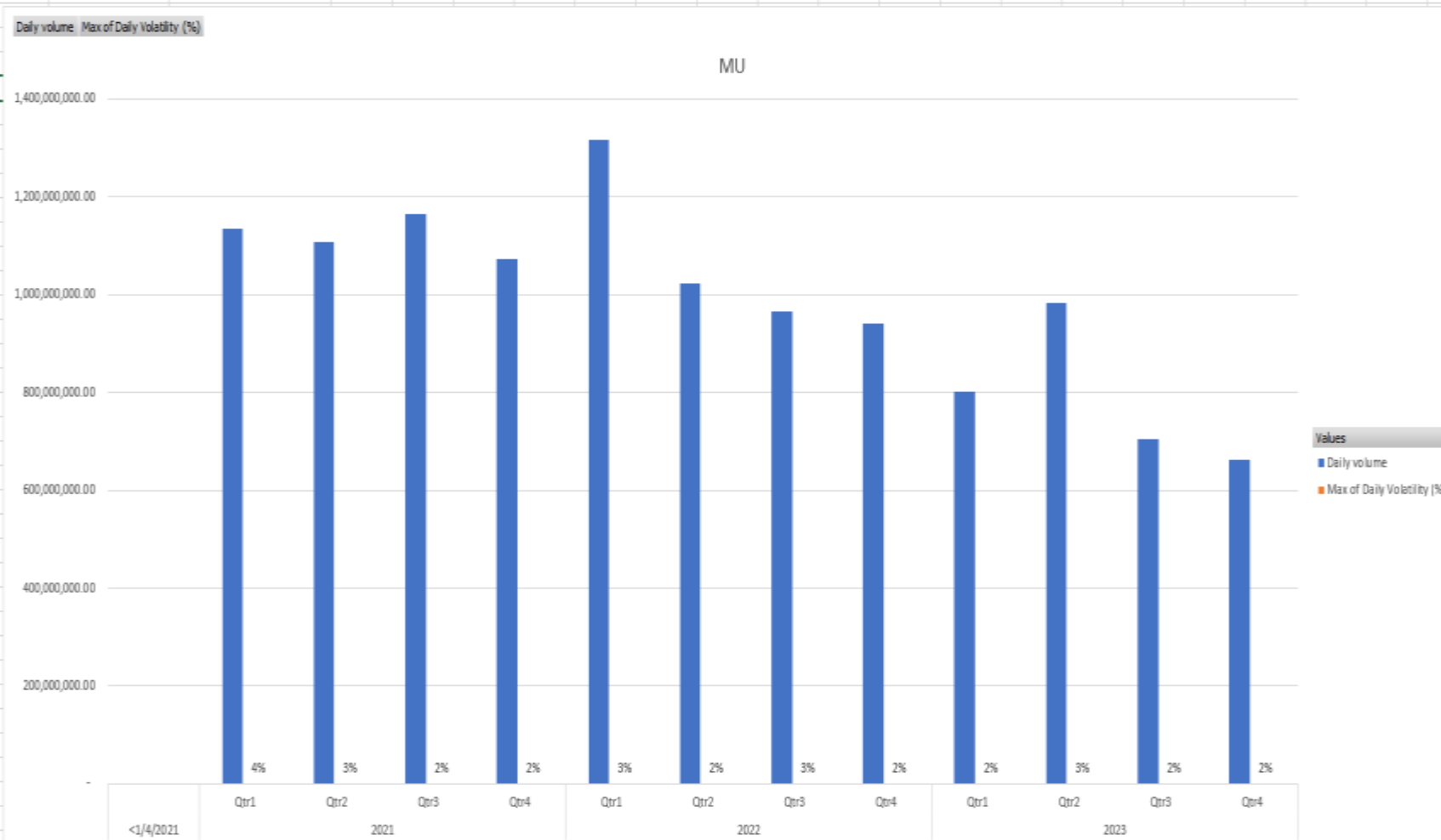
AMD

Row Labels	Daily Volume	Max of Daily Volatility
<1/4/2021		
2021	12,324,189,740.00	5%
2022	21,812,932,749.00	5%
2023	14,311,100,443.00	9%
Grand Total	48,448,222,932.00	9%





Row Labels	Sum of volume	Daily Volatility
<1/4/2021		
2021	4,855,558,432	6.3%
2022	12,686,404,859	7%
2023	10,777,541,231	5.5%
Grand Total	28,319,504,522	7%



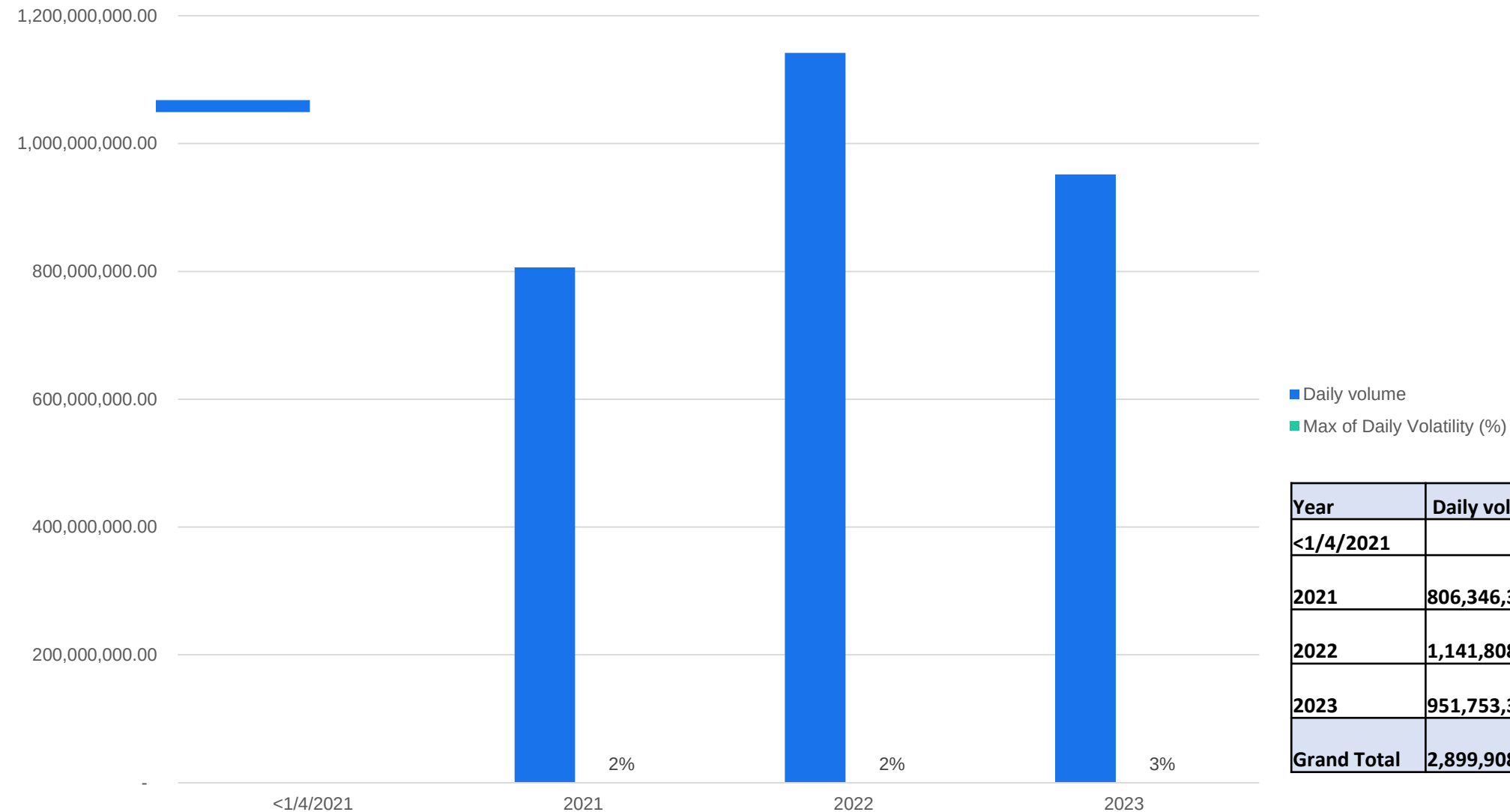
Row Labels	Daily volume	Max of Daily Volatility (%)
<1/4/2021		
2021	4,479,062,671.00	4%
2022	4,245,440,573.00	3%
2023	3,153,829,938.00	3%
Grand Total	11,878,333,182.00	4%

The bars represent MU's total trading volume by quarter from 2021 to 2023. Volumes are:

- Large in every quarter.
- Fairly stable over time (no crazy spike like NVDA 2022, but consistently strong).

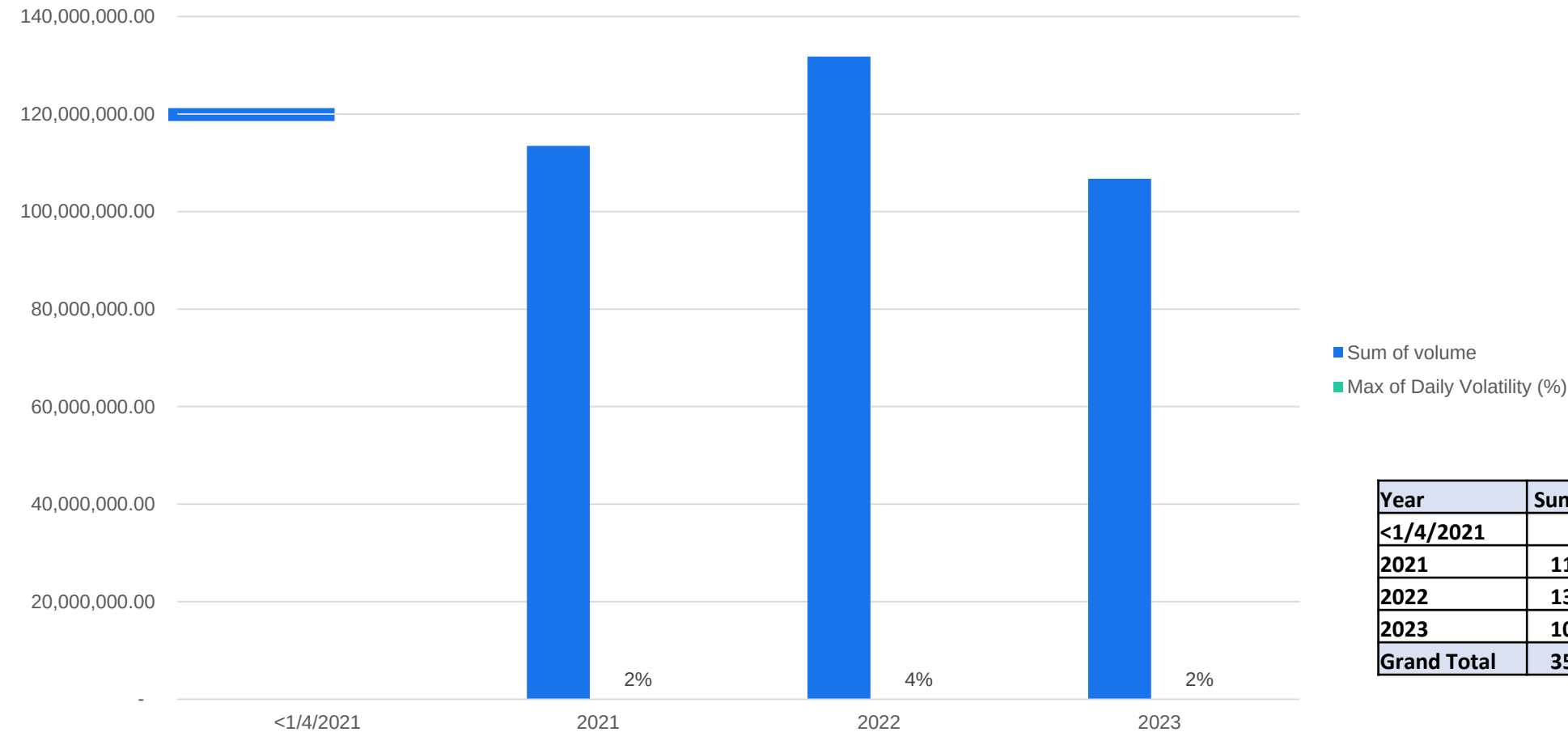
MU trades strong, steady volume across all three years, but its worst single-day intraday move each year is only around 3–4%.

TXN



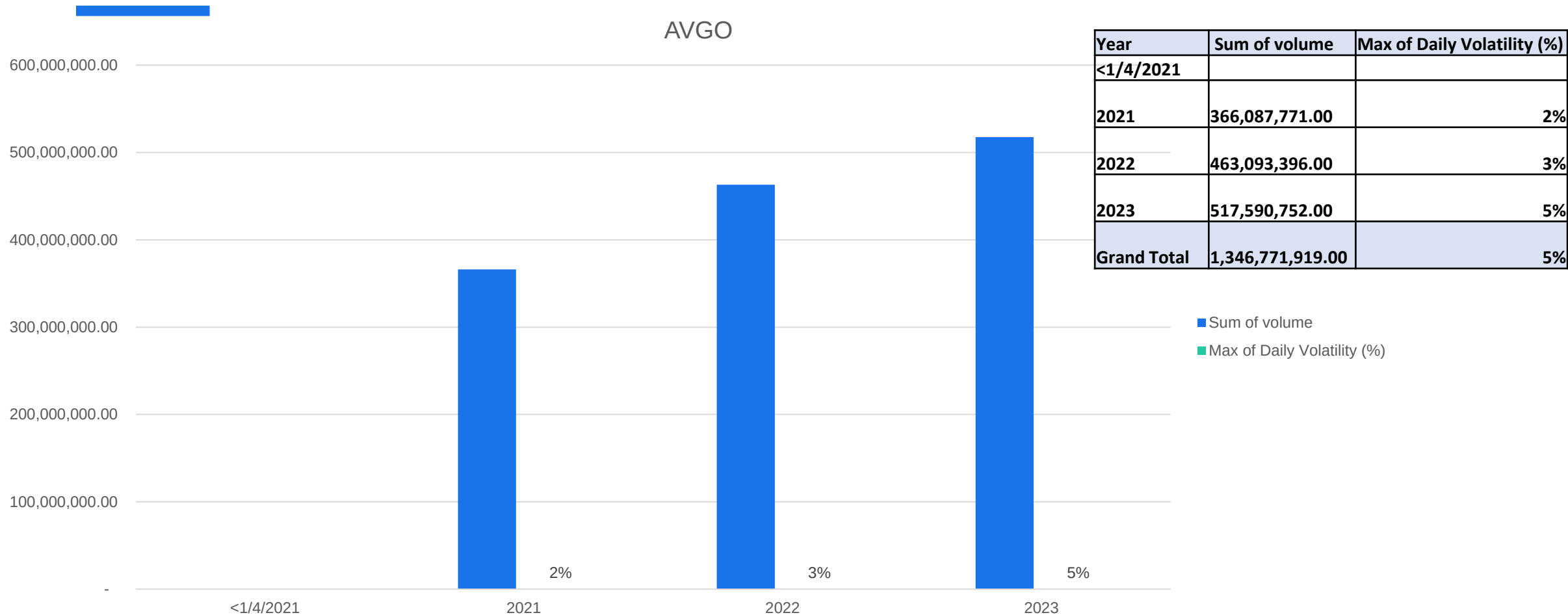
For TXN, annual trading volume stays high and fairly stable from 2021–2023, while the maximum single-day intraday volatility remains low at only 2–3%, making it one of the more stable names in our semiconductor universe

SOXX



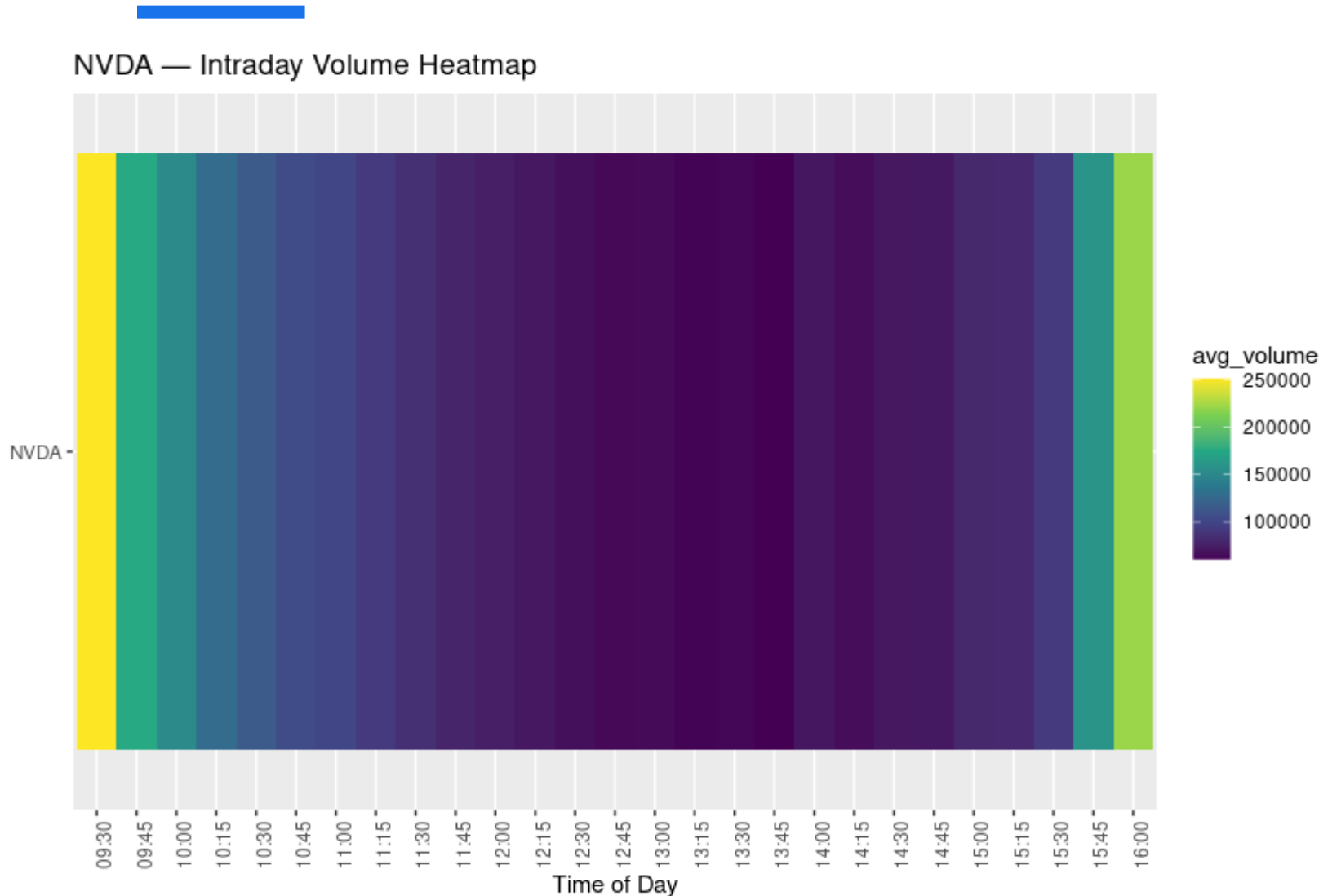
Year	Sum of volume	Max of Daily Volatility (%)
<1/4/2021		
2021	113,455,133.00	2%
2022	131,796,519.00	4%
2023	106,721,057.00	2%
Grand Total	351,972,709.00	4%

The chart summarizes SOXX (the semiconductor ETF) by year :
SOXX trades a lot every year, but its worst single-day intraday move is only around 2–4%, so it's a very liquid and relatively low-volatility way to get semiconductor exposure compared with individual stocks like NVDA or AMD



For AVGO, annual trading volume rises steadily from 2021 to 2023, while the maximum single-day intraday volatility only increases modestly from 2% to 5%, indicating growing activity but still relatively controlled price swings

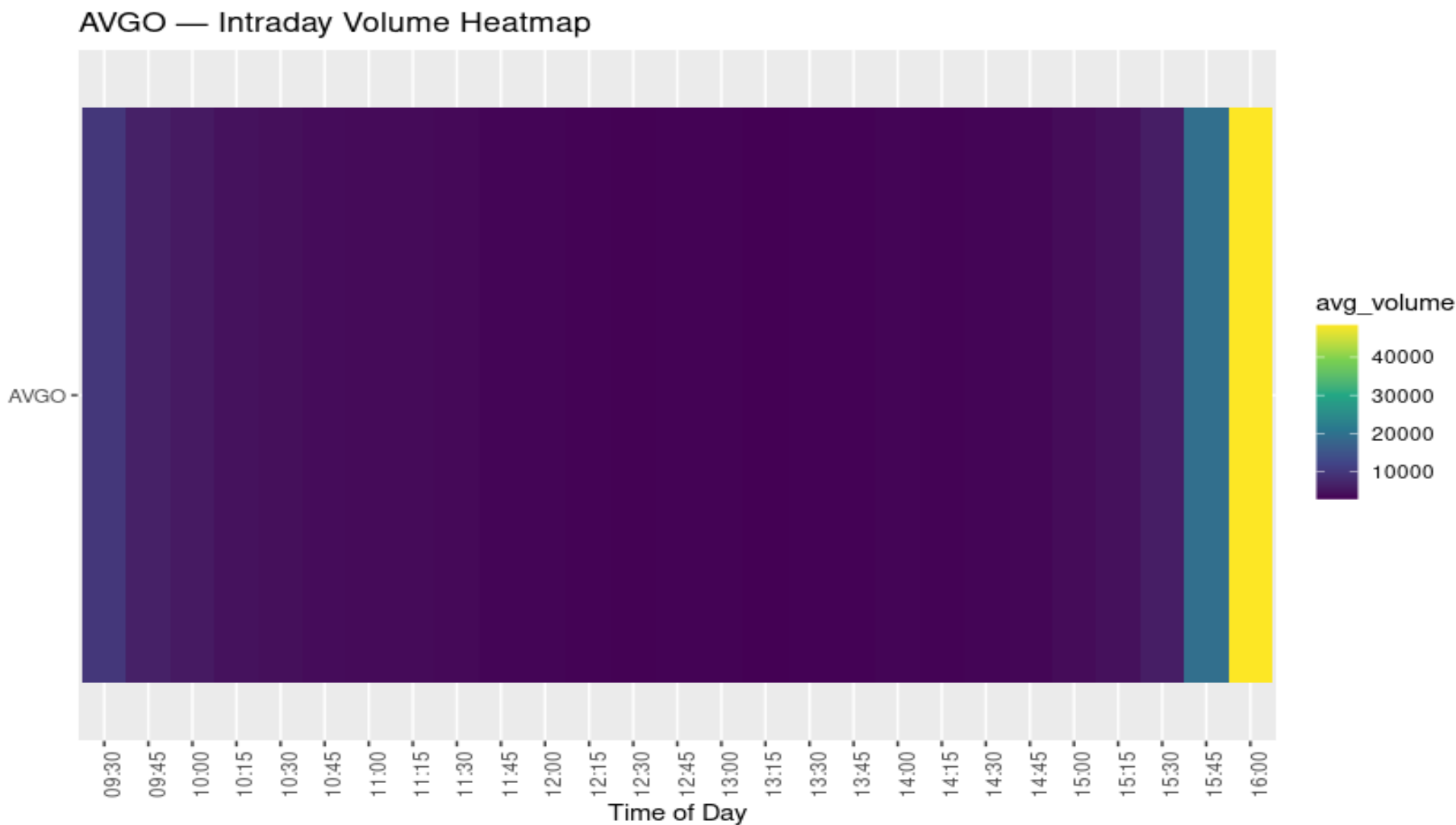
Insight 2 Heat Maps : TOP Volume Stock



“For NVDA, average intraday volume over three years is heavily concentrated at the open and into the close, with a clear lull in the middle of the trading day.

- The heatmap shows a big spike in volume right at the open (09:30).
- Volume then drops off through late morning and early afternoon.
- It ramps back up into the close (around 15:30–16:00)

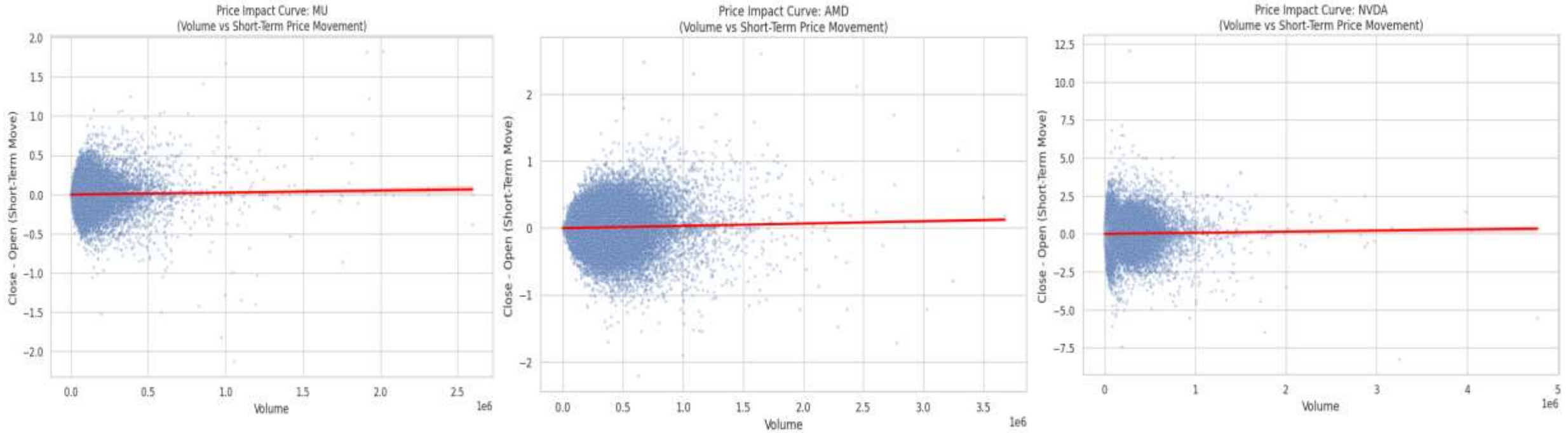
Insight 2 Heat Maps : Bottom Volume Stock



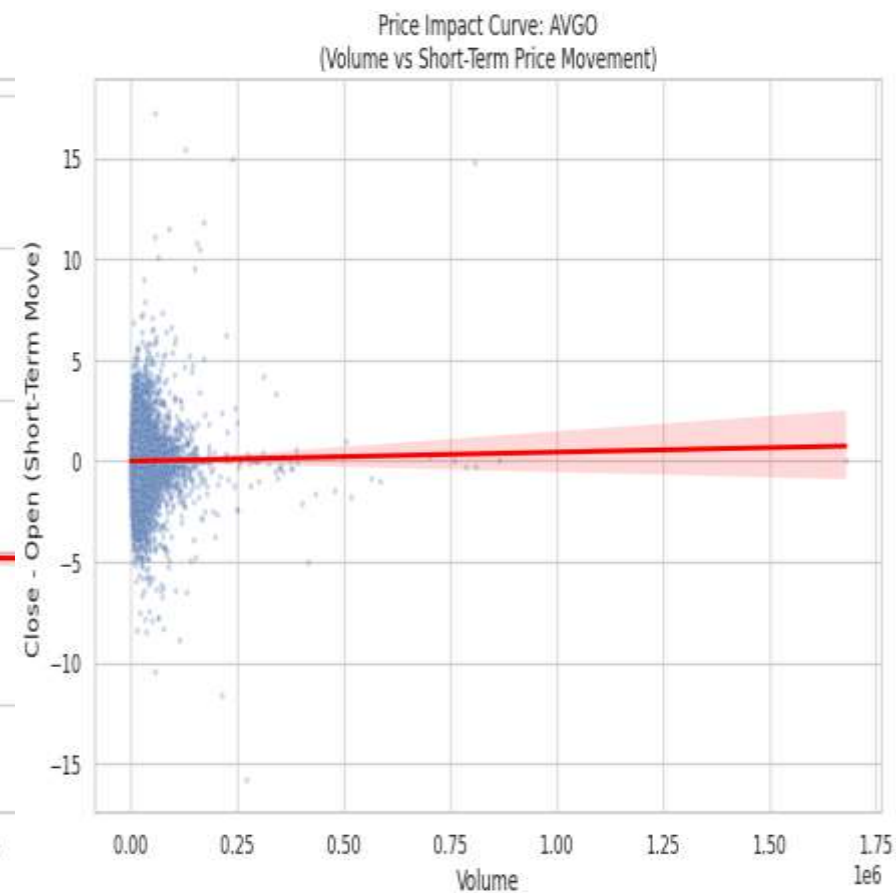
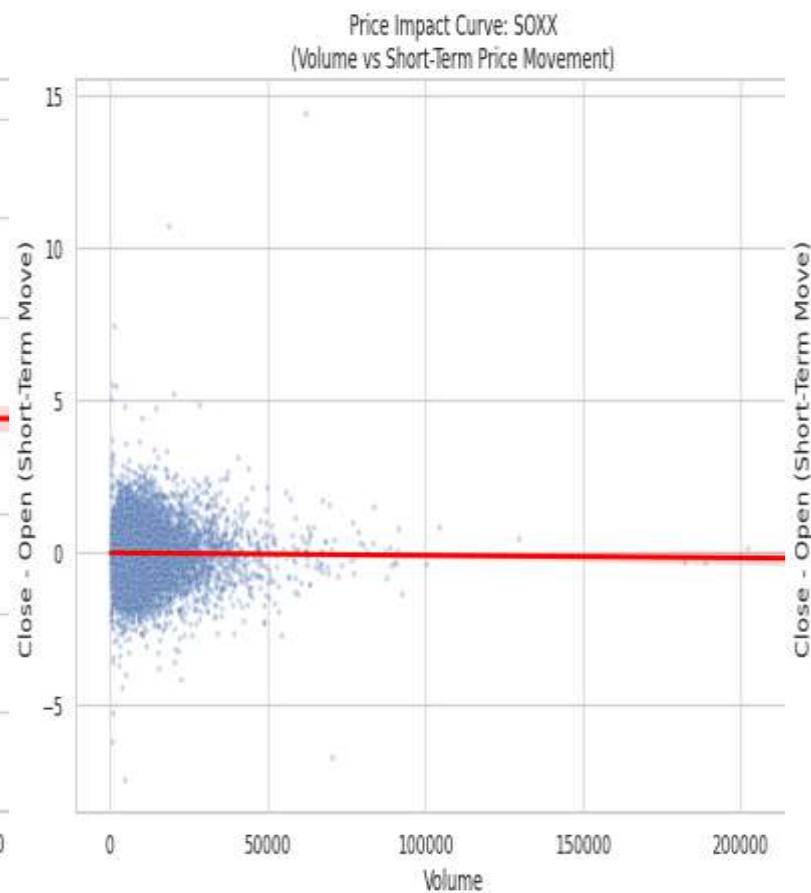
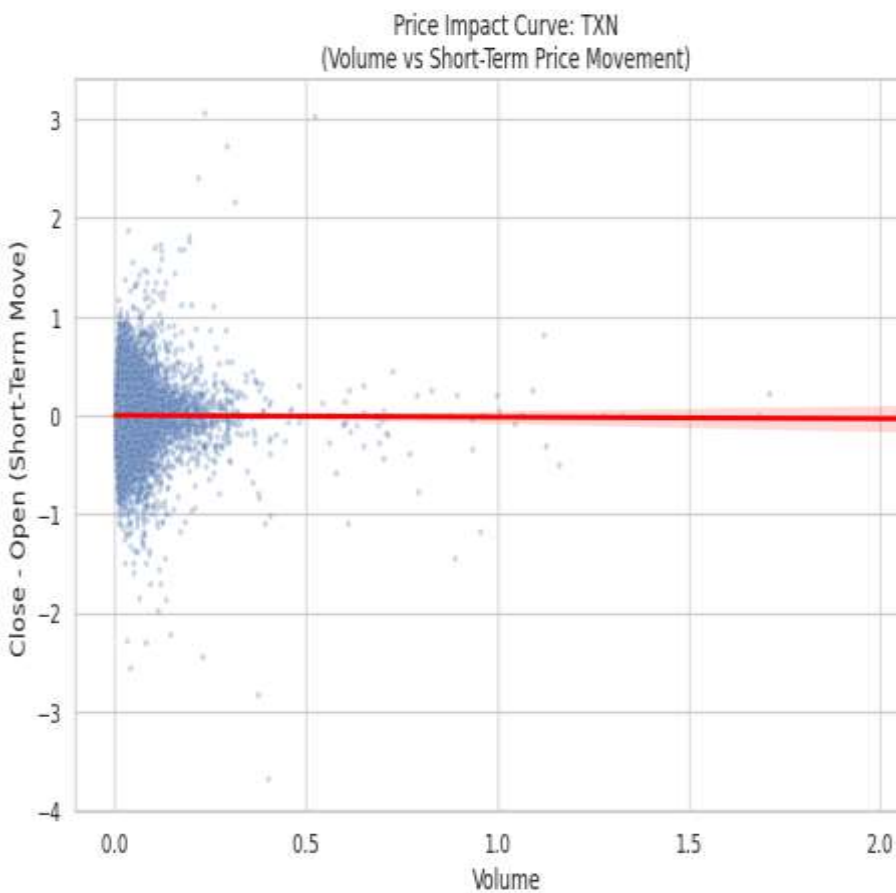
High Volume upon Closing across the 3-year Time Period.

Shows Predictability and potential win opportunity for intraday trades

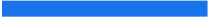
Insight 3 :Price Impact Curves (Volume vs Price)



High Volume stocks – Stock spread contained within 5 units with minimal outliers



Lower Volume Stocks – More spread than Higher Volume Stocks



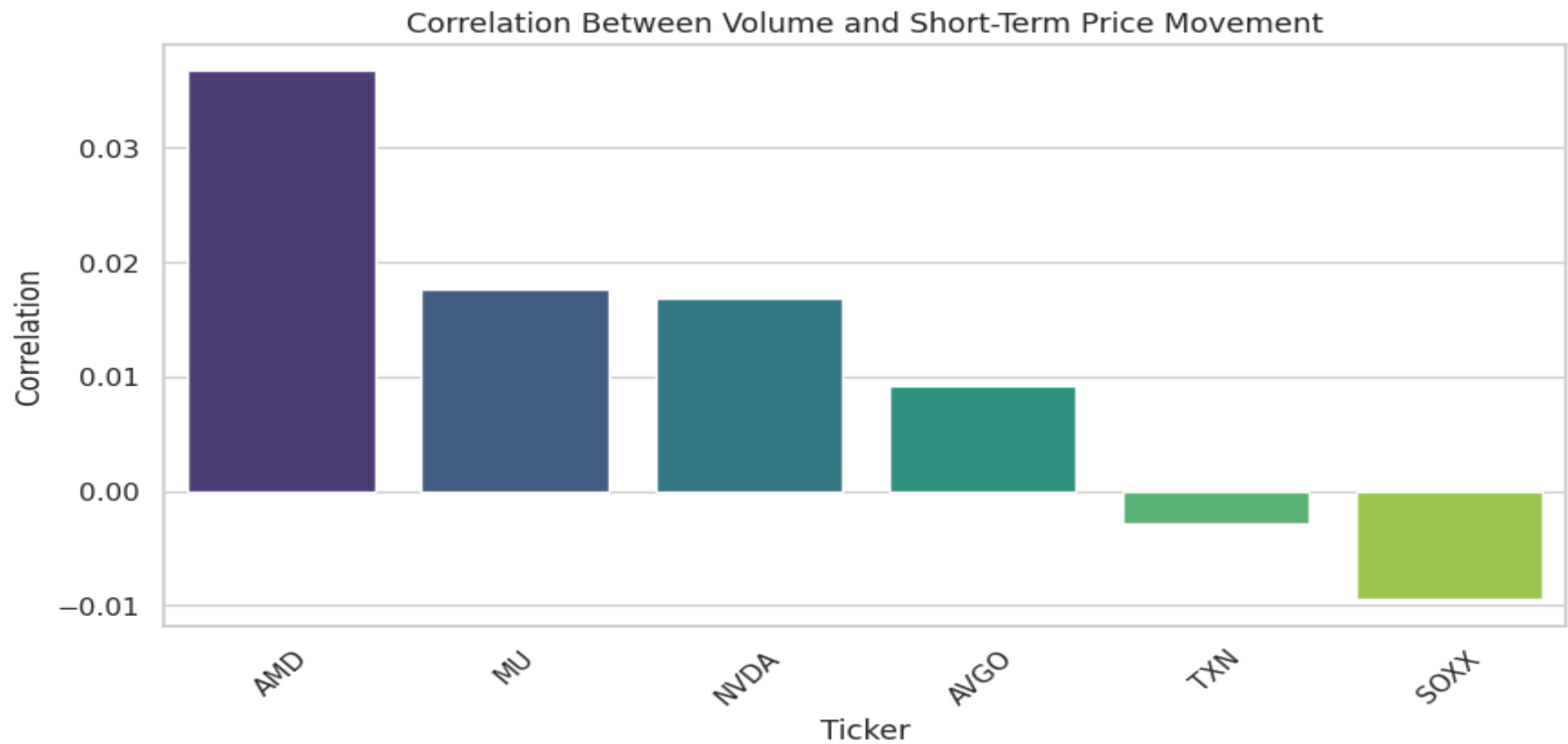
CONCLUSIONS

- 1) Do semiconductor stocks show common market-wide volume–price patterns?
- 2) Which stocks have the most predictable short-term price movements based on volume?

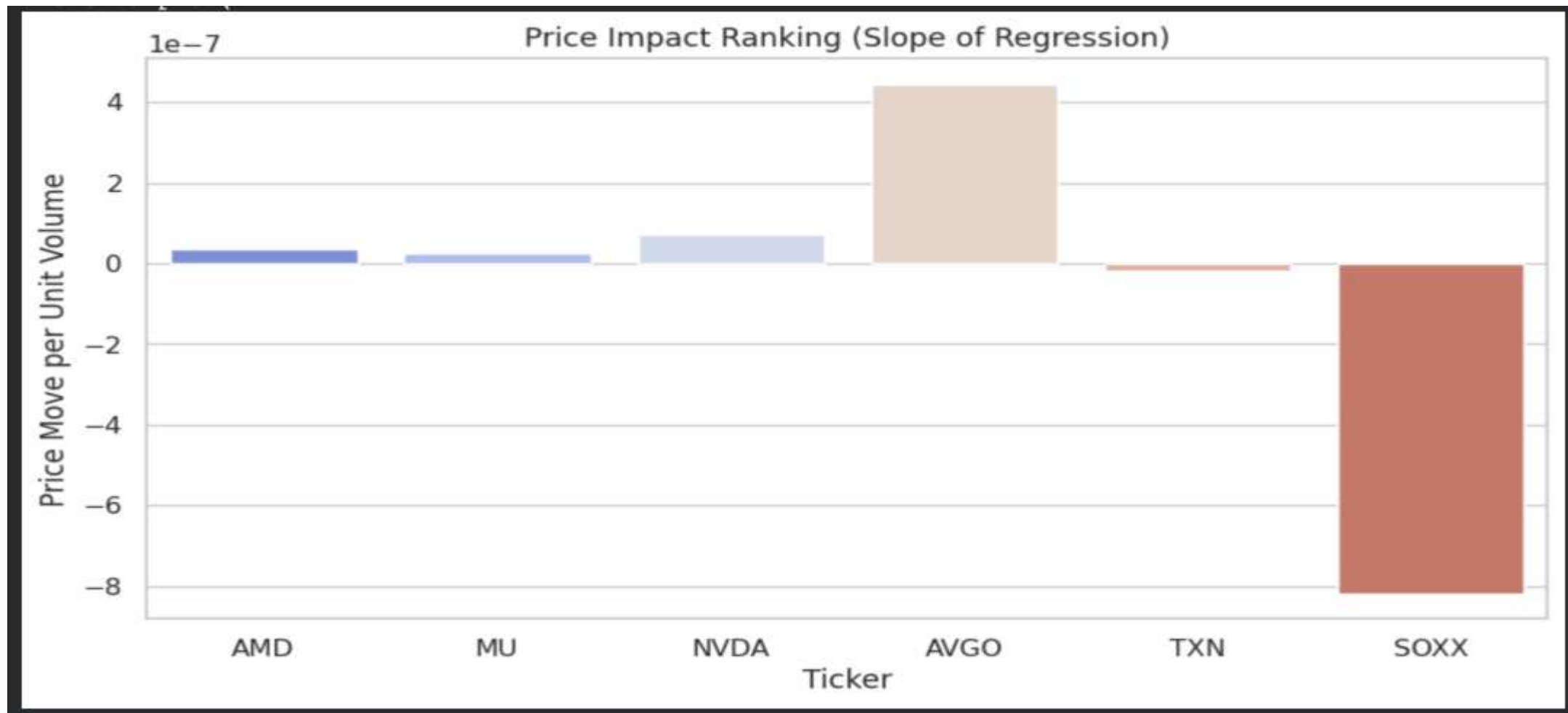


Industry-wide volume does not reliably predict short-term price movement.

The semiconductor sector behaves in a fragmented way, with no common liquidity-driven pattern across stocks.



AMD, MU, and NVDA are the most predictable stocks based on volume, showing consistent positive reactions to trading activity. TXN and SOXX are the least predictable, with SOXX often moving opposite to volume due to ETF flow mechanics.



- **AVGO** shows the strongest positive regression slope (price impact), meaning that even small changes in volume can move its price noticeably.
- **SOXX** again stands out with a **strong negative slope**, reinforcing that large ETF volume often occurs during sell-offs, not rallies.

APPENDIX

SQL CODE FOR DATA EXTRACTION

```
/* SOXX*/  
SELECT  
tickerdate,  
tickertime,  
open,  
high,  
low,  
close,  
vw_price,  
volume,  
trades  
FROM polygon_1m_bars.soxx  
WHERE tickerdate BETWEEN '2021-01-01' AND '2023-12-31'  
AND tickertime BETWEEN '09:30:00' AND '16:00:00'  
ORDER BY tickerdate, tickertime;
```

R STUDIO : Convert date & time into proper formats

```
library(dplyr)
library(lubridate)
```

```
fix_data <- function(df) {
  df %>%
  mutate(
    tickerdate = as.Date(tickerdate),
    tickertime = hms::as_hms(tickertime),
    hour = hour(tickertime),      # Extract hour of day
    minute = minute(tickertime),
    price_change = close - open, # Short-term movement
    volatility = high - low      # Intrabar range
  )
}
```

```
nvda <- fix_data(nvda)
avgo <- fix_data(avgo)
```

R STUDIO Aggregate by time of day across 3 years

```
nvda_summary <- nvda %>%  
  group_by(time_of_day = format(time_bin, "%H:%M")) %>%  
  summarise(  
    avg_volume = mean(volume, na.rm = TRUE),  
    avg_volatility = mean(volatility, na.rm = TRUE)  
  )
```

```
avgo_summary <- avgo %>%  
  group_by(time_of_day = format(time_bin, "%H:%M")) %>%  
  summarise(  
    avg_volume = mean(volume, na.rm = TRUE),  
    avg_volatility = mean(volatility, na.rm = TRUE)  
  )
```

R STUDIO : Create heatmaps in RStudio

(ggplot2)

```
library(ggplot2)
```

```
# NVDA Volume Heatmap
```

```
ggplot(nvda_summary, aes(x = time_of_day, y = "NVDA", fill = avg_volume)) +  
  geom_tile() +  
  scale_fill_viridis_c() +  
  labs(title = "NVDA — Intraday Volume Heatmap",  
        x = "Time of Day", y = "") +  
  theme(axis.text.x = element_text(angle=90, hjust=1))
```